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industry experts. Sivarajan says, "The cost of semiconductor is falling every day, be it microcontrollers, metal-oxide-semiconductor field-effect transistors (MOSFETs) or IGBTs, which impacts the overall pricing of power backup systems."

"We cannot continue using power backup systems with older technology. Prices of products incorporating newer technology will be on the higher side in initial stages, but the advantages these will offer will prove tremendous later," he adds.

Indian inverter market is divided into sine-wave and square-wave inverters. Sood notes, "North India is price conscious and [people here] go for square-wave inverters, which typically cost around ₹ 4000, whereas South India is more technology conscious and [people there] opt for sine-wave inverters, which cost around ₹ 4500."

Powerful future for UPS and inverter industry

The future of the UPS and inverter market, especially in the industrial segment, seems bright. There is traction towards renewable energy systems like solar inverters and converters. Sivarajan feels, "In the coming years, we must shift our dependency to solar instead of generating more power from non-renewable sources."

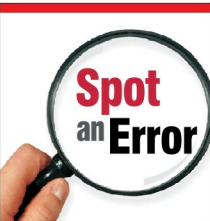
Bureau of Indian Standards (BIS) and Bureau of Energy Efficiency (BEE) standards, which focus on safety and efficiency of inverters, are still to be implemented. Sood says, "With BEE standards, demand will

slowly decline for square-wave inverters. Also, with new government policies and subsidies being offered, solar systems will be gaining momentum in the market."

She adds, "Renewable systems see a very huge growth potential in the corporate inverters market. Although inverter companies are already working towards this solution, there should be policies in place for encouraging development of inverters that will feed back power to the grid."

The per capita consumption of electricity in India is about 1000kWh and the worldwide consumption is 2600kWh. Even as power generation increases, demand is bound to increase in India, feels Sood. She says, "As power conditions improve, we do not foresee that electricity will be available 24 hours in near future. We will also see more people using hybrid inverters, which are a combination of solar and grid power to provide optimal use." She adds, "Inverters in future will not only help create backup for devices but will also be smart enough in order to manage power."

Sivarajan feels, the future will move towards direct current (DC) systems. "Traditional inverters converting DC to alternating current (AC) have their own conversion losses. Complete home lighting and fan application will be driven towards DC for the higher efficiency and safety these provide." He adds, "Lighting will be on light emitting diodes (LEDs), which run on DC and we will have DC fans in ceilings." ●



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